INTRODUCTION TO THE ANALYSIS OF TIME SERIES

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Course contents:

- ☐ Introduction to the analysis of time series
  a. Definition
  b. Components of a time series
  c. Decomposition of a time series on components
  d. Exponential smoothing techniques

- ☐ Linear autoregressive models
  a. Stationary time series, non-stationary time series and non-homogenous time series
  b. Classification of non-stationary time series
  c. Lag operators
  d. Time series characteristics, autocorrelation function and partial autocorrelation function
  e. Non-Stationarity testing (unit roots tests)
  f. AR(p), MA(q), ARMA(p,q), ARIMA(p,d,q) models: definition and main characteristics
  g. Box Jenkins methodology: stationarity analysis, transformations into a stationary time series, model identification, estimation of parameters, goodness of fit tests, selection the best model given the data, predictions or forecasts

- ☐ Advanced ARIMA models:
  a. Autoregressive moving average models with seasonal evolution - SARIMA,
  b. ARCH, GARCH models.

- ☐ Multivariate stationary time series models
  a. VAR models (Vector AutoRegression models)
  b. Granger causality

- ☐ Multivariate non-stationary time series models
  a. Cointegration
  b. VECM models (Vector Error Correction models)
  c. Cointegration tests

Final grade

- Written final exam 70%
- Active participation in seminar classes (attendance, questions and answers) 10%
- Term project to be presented in seminar 20%

References

1. Peter J. Brockwell, Richard A. Davies. *Introduction to Time Series and Forecasting*, Springer, Second Edition, 2002, Third Edition 2016.
2. Wei, W. W. S., *Time Series Analysis – Univariate and Multivariate Methods*, Second Edition, Pearson Addison Wesley, 2006.
3. Ruey S. Tsay. *Analysis of Financial Time Series*, Wiley, Second Edition, 2010.
4. Damodar N. Gujarati, Dawn C. Porter. *Basic Econometrics*, 5th ed, McGraw-Hill, 2009.
5. Enders, W., *Applied Econometric Time Series*, Wiley, 4th ed, 2015, companion website https://bcs.wiley.com/he-bcs/Books?action=index&itemId=1118808568&bcsId=9288, https://bcs.wiley.com/he-bcs/Books?action=index&itemId=1118808568&bcsId=9265, http://time-series.net .
6. EViews software www.eviews.com , EViews Online Tutorials https://eviews.com/Learning/index.html
7. Pfaff, B., *Analysis of Integrated and Cointegrated Time Series with* R, Springer, 2008.
8. Hyndman, R. J., Athanasopoulos, G., *Forecasting: Principles and Practice* (3rd ed), available at https://otexts.com/fpp3/
9. Shumway, R. H., Stoffer, D. S., *Time Series Analysis and Its Applications With* R *Examples*, Fourth Edition, Springer, 2017.
10. Andrei, T., Stancu, S., Iacob, A., I., Tusa, E, *Introducere în econometrie utilizând EViews*, Editura Economică, Bucureşti, 2008, România.
11. Andrei, T., Oancea, B., Mirică, A., Toma, I. E., Herțeliu, C., Econometrie. *Teorie și aplicații în EViews și* R, Editura Economică, Bucureşti, 2018, România.
12. Voineagu V., Ţiţan E., Şerban R., Ghită S., Todose D., Boboc C., Pele D., *Teorie şi Practică Econometrică*, Meteor Press, Bucureşti, 2007, România.

Some statistical data sources for time series (of course, the list is not exhaustive)

- Institutul Național de Statistică http://www.insse.ro/cms/ (TEMPO Online)
- Banca Națională a României http://www.bnr.ro/Home.aspx
- World Bank http://www.worldbank.org/
- Eurostat http://ec.europa.eu/eurostat
- Federal Reserve Bank of St. Louis https://fred.stlouisfed.org/
- The University of York, Department of Mathematics https://www.york.ac.uk/depts/maths/data/ts/
- Yahoo Finance https://finance.yahoo.com/
- https://demo.quandl.com/
- https://www.investing.com/

Software:

- **EViews**, https://www.eviews.com/home.html, EViews 12 Student Version Lite can be freely downloaded from https://www.eviews.com/EViews12/EViews12Univ/evuniv12.html using your student email ,
- **R** – open source software, https://www.r-project.org/ , with **RStudio Desktop** (open source) https://rstudio.com/, https://rstudio.com/products/rstudio/

TIME SERIES – part 1

Let us suppose that someone is interested in the evolution of a variable *Y* over time.

Such a variable *Y* can be:

- GDP,
- Unemployment rate,
- Inflation rate,
- Imports,
- Exports,
- Production,
- Investments,
- Interest rates,
- Exchange rates etc.

The variable is observed at different, but regular, periods of time, and the time between two observations is constant, namely daily, monthly, quarterly, yearly.

Definition: A real-valued discrete **stochastic process** is a sequence of random variables indexed by $t \in \mathbf{Z}$ on the probability space (Ω, K, P)

$$Y(\cdot,\cdot): \mathbf{Z} \times \Omega \to \mathbf{R}$$

$$(t, \omega) \mapsto Y(t, \omega) \overset{not}{=} Y_t(\omega)$$

where Ω is the sample space, $\omega \in \Omega$ is a state of the nature such that $y_t = Y_t(\omega)$, K is a σ-algebra, and P is a probability measure. (Florian Pelgrin)
In short, a stochastic process is a collection of random variables ordered in time, that is $\{Y_t\}_{t \in \mathbf{Z}}$.

Definition: A realization of the stochastic process $\{Y_t\}_{t \in \mathbf{Z}}$ is **a time series** or **a chronological series**, and could be denoted $\{y_t\}_{t \in \mathbf{Z}}$.

Remark: For the sake of notation, no difference is made between the random variables and their realizations.

Remark: A time series is a set of observations $\{y_t\}_{t \in T_0}$ on the variable of interest *Y*, each observation being recorded at a specific time *t*.
A **discrete-time time series** is one in which the set of times at which observations are made is a discrete set $T_0 \subseteq \mathbf{Z}$, as in the case, for example, when observations are made at fixed time intervals.
A **continuous-time time series** is obtained when observations are recorded continuously over some time interval $T_0 \subseteq \mathbf{R}$ (Brockwell, page 1-2).

Remark: The statistical data recorded with its time of occurrence is called a **time series**.
http://www.emathzone.com/tutorials/basic-statistics/introduction-to-time-series.html

Y – quantitative random variable of interest,
real values

$\{Y_0, Y_1, Y_2, \ldots\ldots, Y_{n-1}, Y_n, Y_{n+1}, \ldots\ldots\}$

is a sequence of random variables

is the stochastic process $\{Y_t\}_{t \geq 0}$
$t = 0, 1, 2, \ldots\ldots$

Y_t – r.v. characterizing a statistical unit
at time moment t
or for a time period t

Y – GDP, in millions \$US, Romania 1960

$\{Y_0, Y_1, Y_2 \ldots\ldots\ldots\ldots, Y_{61}, \ldots\ldots\}$ sequence of r.v.

1960 1961 1962 2020

$\underbrace{\{y_0, y_1, y_2, \ldots\ldots\ldots\ldots, y_{61}\}}_{\text{a time series of } \{Y_t\}_t \text{ GDP of Romania}}$ the set of observed values of the above sequence on a given (specified) time horizon

Y – amount of time of winning Tokyo marathon
minutes, second, milliseconds

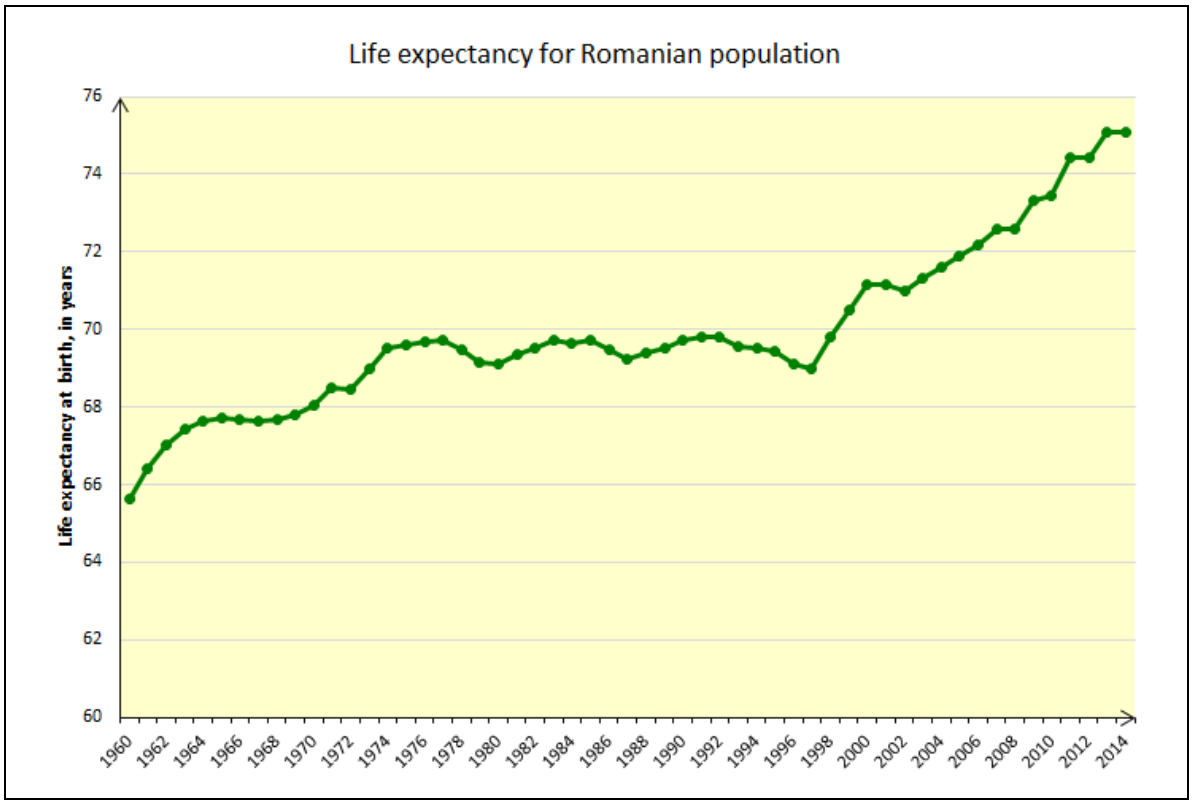
$Y_{\boxed{1975}}, Y_{\boxed{1976}}, \ldots\ldots\ldots, Y_{\boxed{2021}}$ $(Y_t)_{t \text{ discrete}}$

$y_{1975}, y_{1976}, \ldots\ldots, y_{2021}$

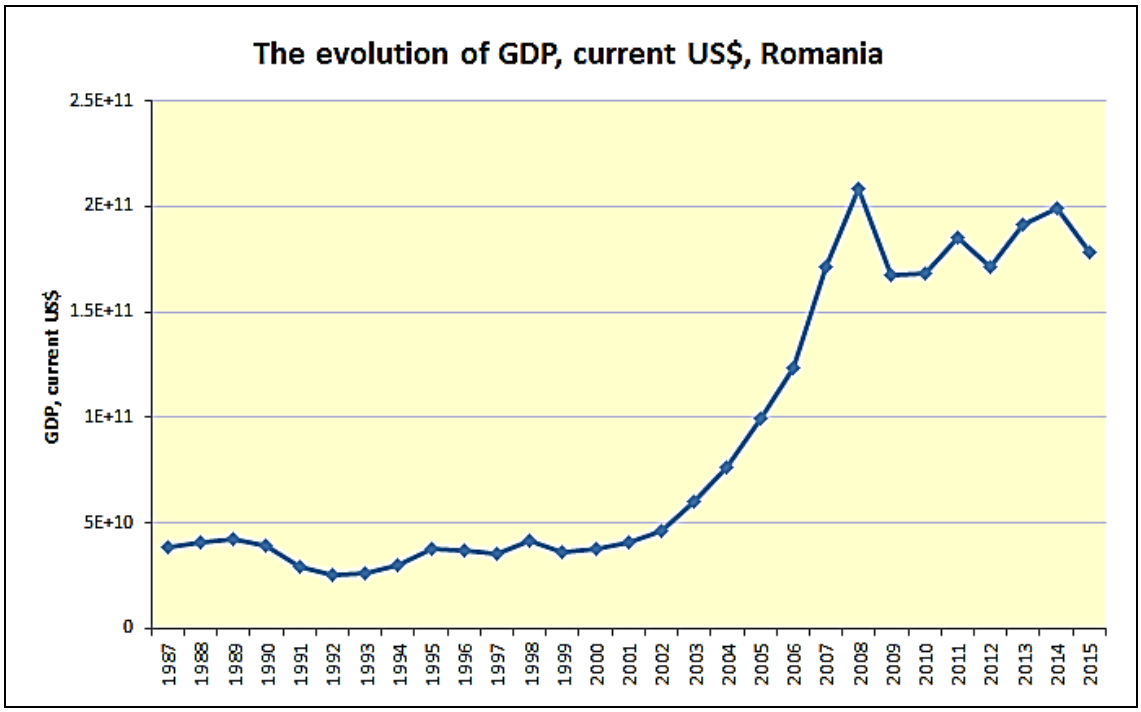
$\{\text{Tesla Price}_t\}_{t = \text{minutes}}$

Examples of time series and **time series plots** (a graph that displays the observed values in the order in which they occurred in time)

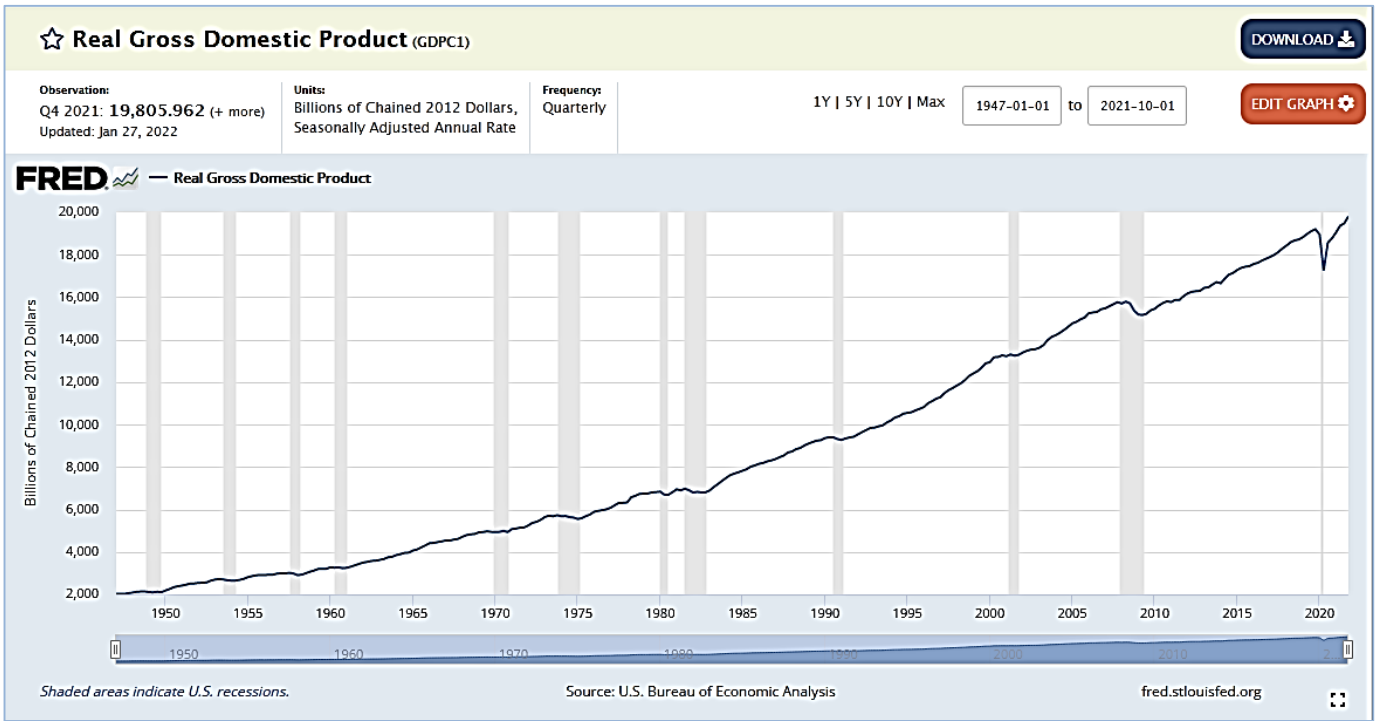
- **Life expectancy at birth, total (years), yearly data**;
  Life expectancy at birth indicates the number of years a newborn infant would live if prevailing patterns of mortality at the time of its birth were to stay the same throughout its life.
  http://www.worldbank.org/



- **GDP, current US$, for Romania (yearly data)** http://www.worldbank.org/

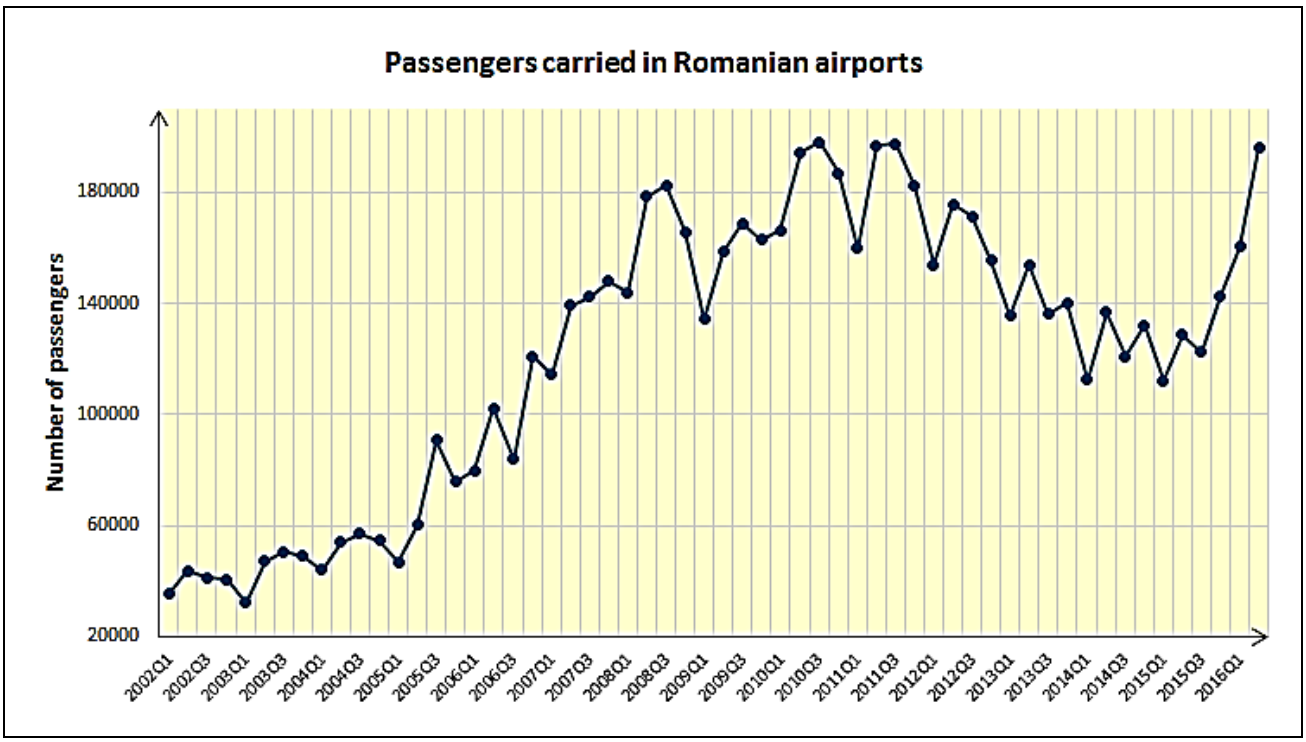


- **Real Gross Domestic Product of US, quarterly data,** https://fred.stlouisfed.org/series/GDPC1

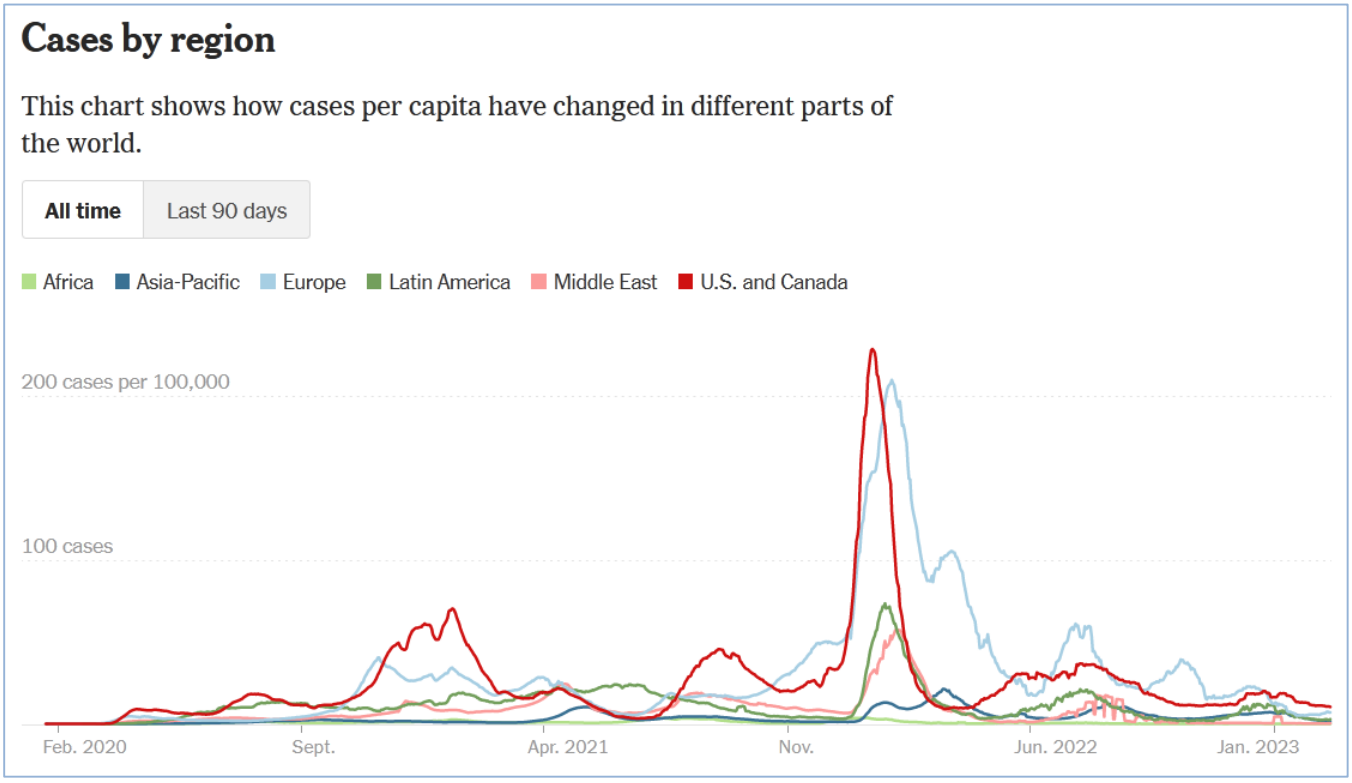


- **The exchange rate GBP (Great Britain Pound) to EUR (daily data)** https://www.xe.com/currencycharts/?from=GBP&to=EUR&view=10Y

- **Number of passengers carried in Romanian airports (quarterly data)**
  http://appsso.eurostat.ec.europa.eu/nui/show.do?dataset=avia_panc&lang=en

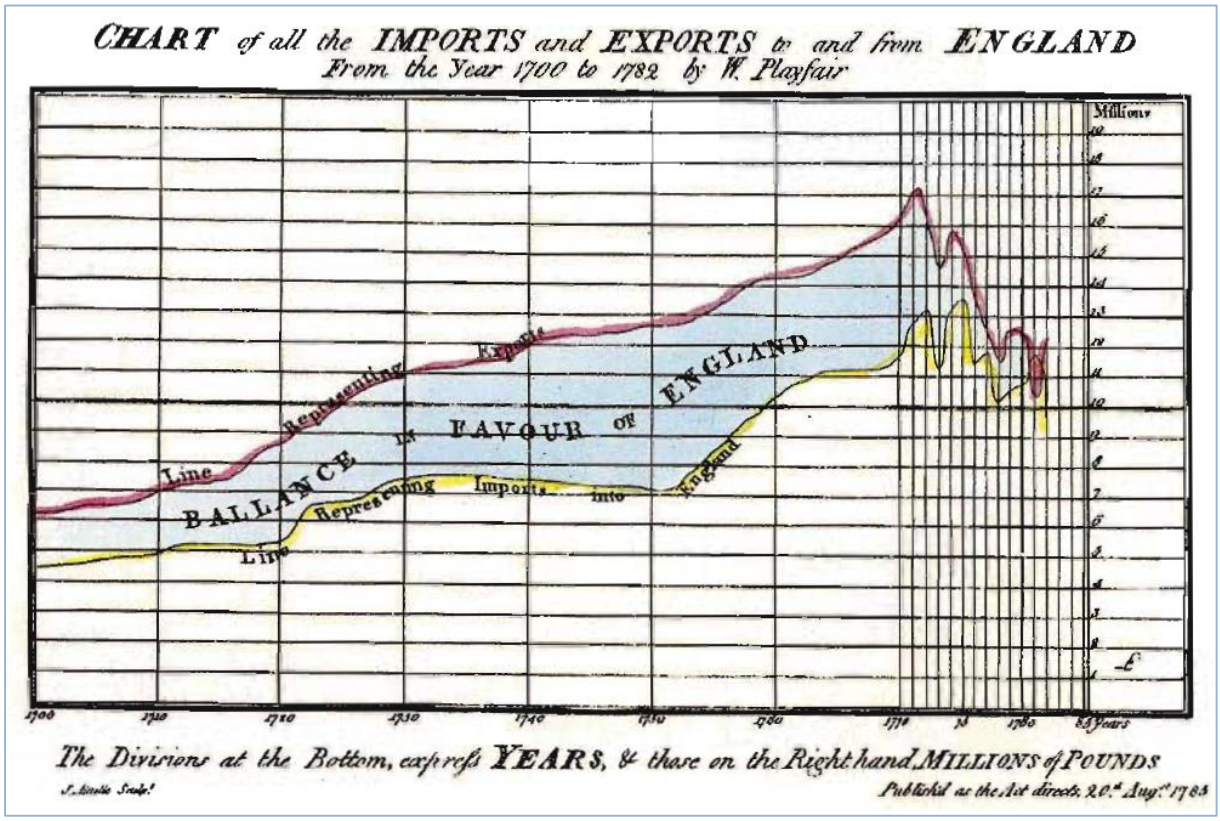


- Number of Covid cases worldwide
  https://www.nytimes.com/interactive/2021/world/covid-cases.html



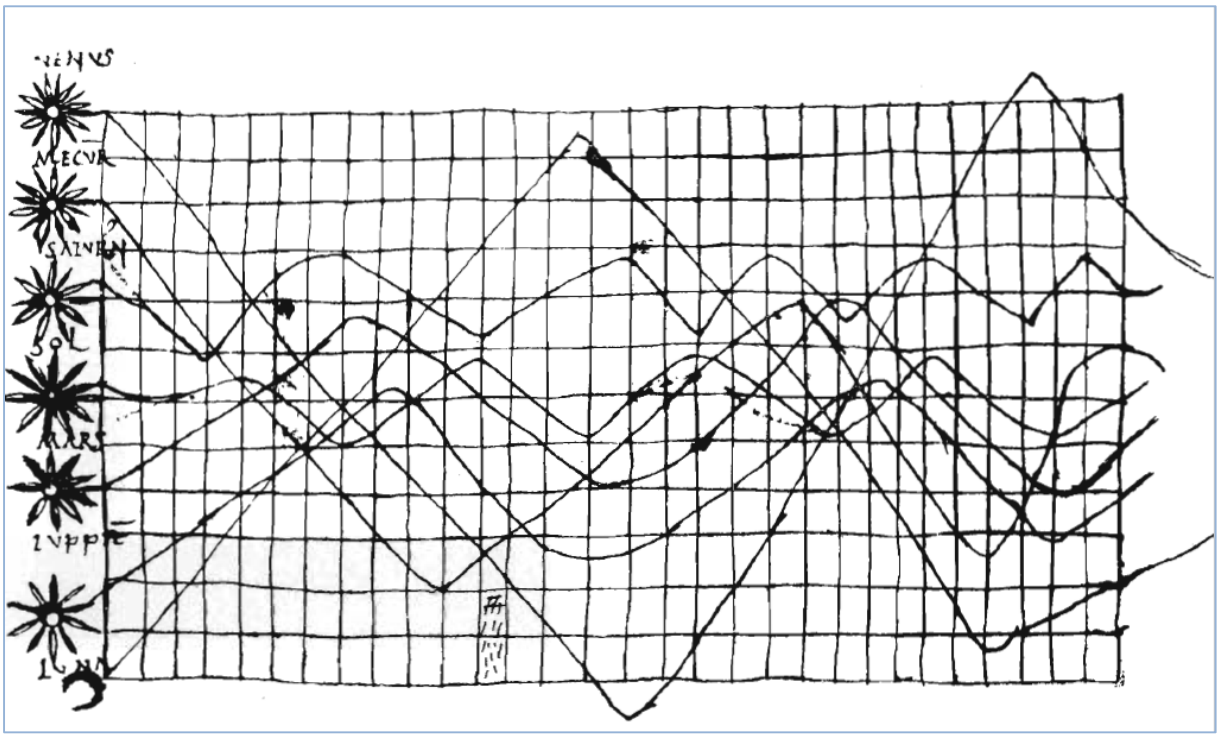
A bit of history in time series plots

According to Tufte[1], page 32, for time series plots, "the two great inventors of modern graphical designs were J. H. Lambert (1728-1777), a Swiss-German scientist and mathematician, and William Playfair (1759-1823), a Scottish political economist. The first known time-series using economic data was published in Playfair's remarkable book, *The Commercial and Political Atlas* (London, 1786)". The time plot below displays the balance of trade for England, as the difference between import and export time series. "Playfair contrasted his new graphical method with the tabular presentation of data."



However, the oldest known time series plot seems to be the plot below which shows "a tenth or possibly eleventh century illustration of the inclinations of the planetary orbits as a function of time, apparently part of a text for monastery schools, is the oldest known example of an attempts to show changing values graphically. It appears as a mysterious an isolated wonder in the history of data graphics, since the next extant graphic of a plotted time-series shows up some 800 years later." (Tufte, page 28).

[1] Edward Tufte, *The Visual Display of Quantitative Information*, second edition, Graphics Press, 2007.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This part of the lecture follows Heinz Kohler, *Statistics for Business and Economics*, Scott, Foresman and Company, 1988.

Time-series analysis: a statistical procedure that employs time-series data, usually for the purpose of explaining past or making forecast for the future. (Heinz Kohler)

Several models seek to describe the typical time series.

The classical time-series model attempts to explain the pattern observed in an actual time series by the presence of four components: trend, cyclical, seasonal, and irregular components.

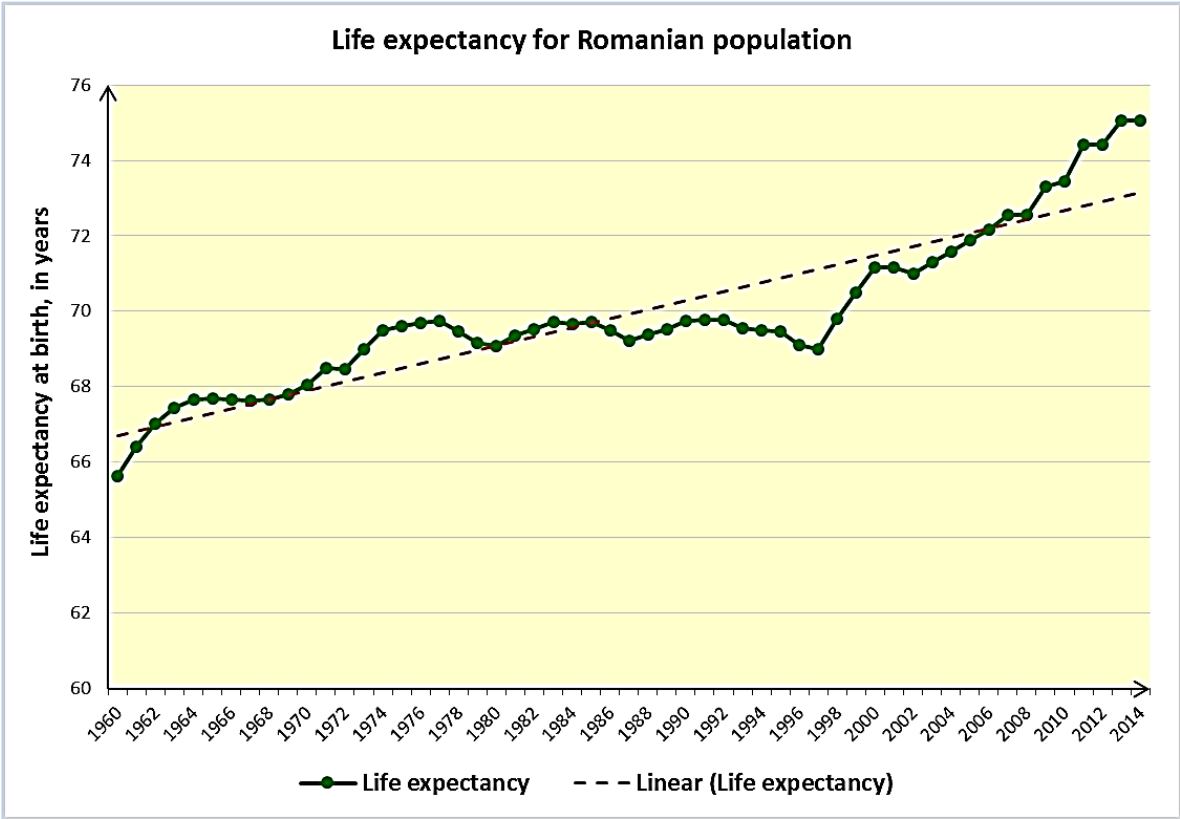
Let us denote by *Y* the random variable of interest
and let $\{y_t\}_{t=\overline{0,n-1} \text{ or } t=\overline{1,n}}$ be the observed time series.

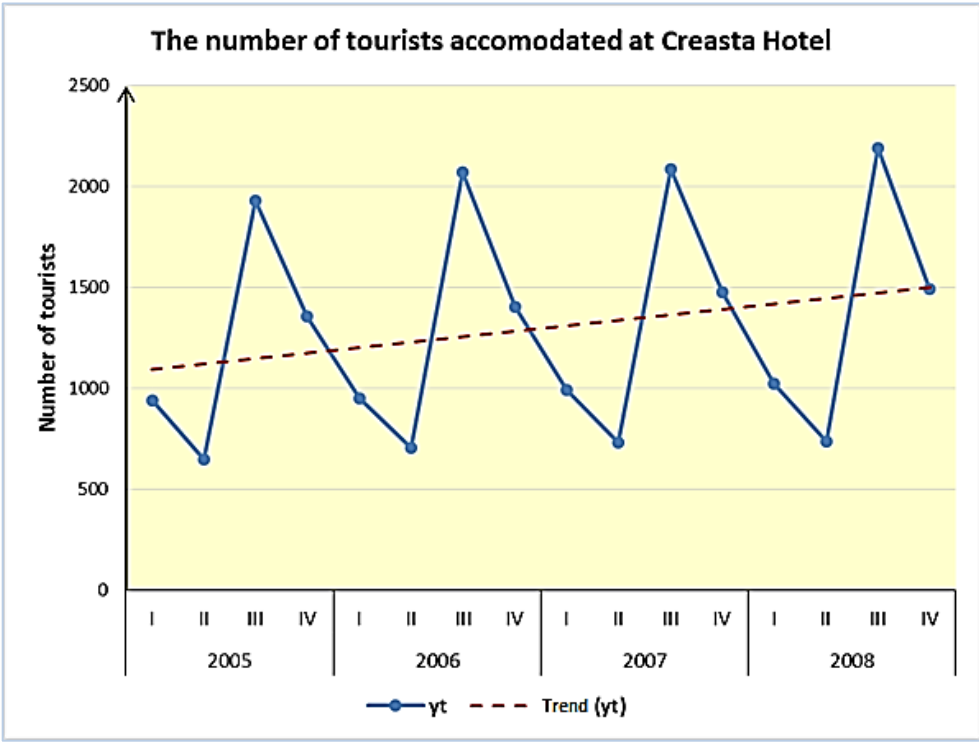
THE COMPONENTS OF A TIME SERIES

A primary use of time-ordered data is to detect trends and other patterns that occur over time.

THE TREND COMPONENT

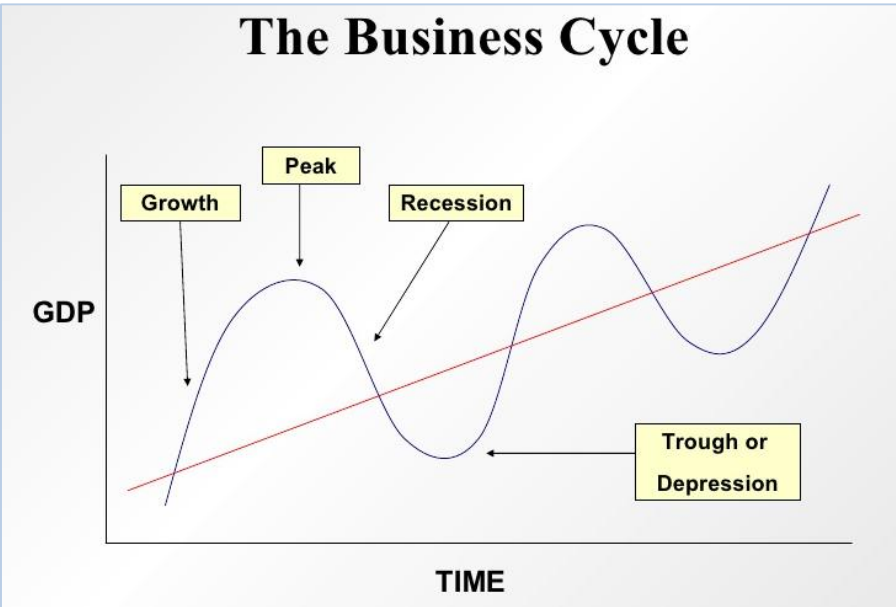
- The trend is the long-term pattern of a time series.
- It is the main component of a time series which results from long-term effect of socio-economic and political factors
- The trend may show the growth or decline in a time series over a long period of time.
- The form of the trend may be linear or non-linear
- It is denoted by y_{tT} and it is evidenced as a relatively smooth, progressively upward or downward movement of the variable of interest Y over an extended period of time





THE CYCLICAL COMPONENT

- The long-term oscillations occurring in a time series
- It is denoted y_{tC} and it is evidenced by wide up-and-down swings of the variable of interest around the trend, with the swings lasting from one to several years each and typically differing in length and amplitude from one cycle to the next
- These oscillations are associated to the well-known business cycles

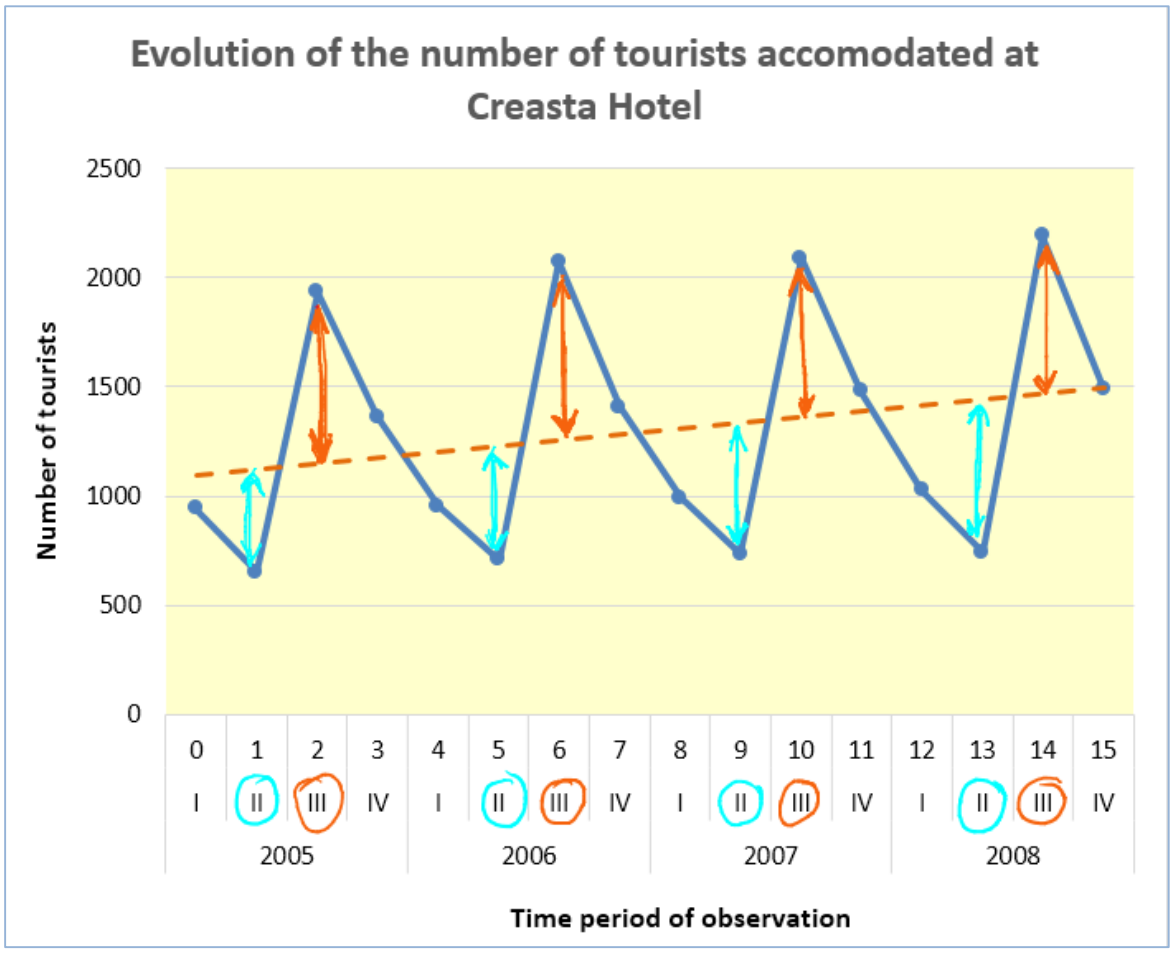


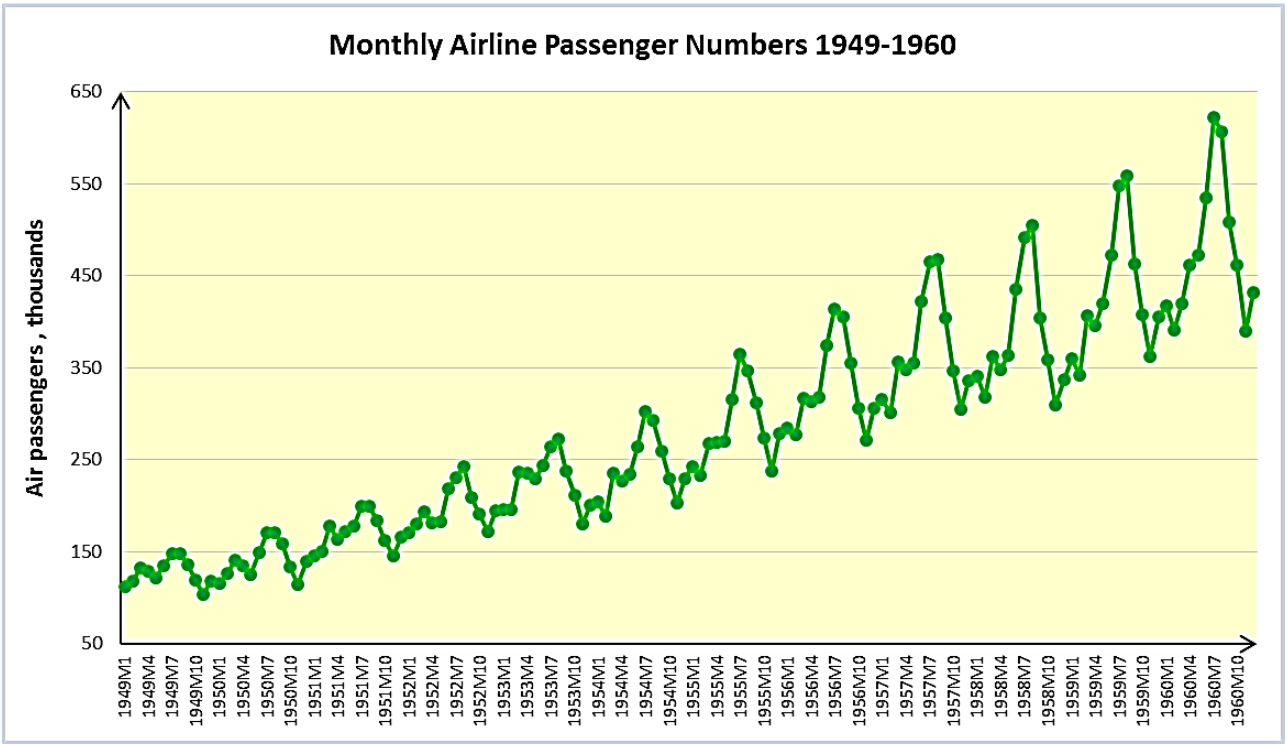
https://www.slideshare.net/henrjt/the-business-cycle-299518

- The duration of a cycle depends on the type of business or industry being analyzed (in economics data the periods of such oscillations are generally extended from five to twelve years or more)

THE SEASONAL COMPONENT

- Short-time movements occurring in data due to seasonal factors
- These types of variations in a time series can only be isolated when the series is provided biannually, quarterly or monthly
- It is denoted y_{tS} and it is evidenced by narrow up and down swings of the variable of interest around the trend component or cyclical component, with swings predictably repeating each other with periods of one year or less
- Regularly recurring intrayear variations often reflect the influence of the weather and the calendar of economic activity





AirPassengers data – The classic Box & Jenkins airline data. Monthly totals of international airline passengers, in thousands, 1949 to 1960. **Source** Box, G. E. P., Jenkins, G. M. and Reinsel, G. C. (1976) *Time Series Analysis, Forecasting and Control.* Third Edition. Holden-Day. Series G.

THE IRREGULAR COMPONENT

- It is denoted y_{tR} and it is evidenced by random movements (fluctuations) of the variable of interest around the trend or cyclical or seasonal component
- This component is unpredictable
- Sudden changes occurring in a time series which are unlikely to be repeated; such movements are viewed as arising from completely unpredictable and probably nonrecurring chance events, such as strikes, fires, floods, natural disasters, wars etc.
- It is that component of a time series which cannot be explained by trend, cyclic or seasonal components
- It is also called *residual component*, and can be also denoted ε_t .

Given the time series $\{y_t\}_t$, how do the above components combine to result in a value of y_t ?

The classical additive time-series model:

$$\underbrace{y_t}_{\text{data}} = \underbrace{y_{tT} + y_{tC} + y_{tS}}_{\text{predictable}} + \underbrace{y_{tR}}_{\text{residual}}$$

The classical multiplicative time-series model:

$$\underbrace{y_t}_{\text{data}} = \underbrace{y_{tT} \times y_{tC} \times y_{tS}}_{\text{predictable}} \times \underbrace{y_{tR}}_{\text{residual}}$$